10. Ejercicios

Ejercicio 10.1. Integración directa

1. $\int (x^8 - 3x^4 + 2)dx.$
2. $\int (4x^{-5} + 3\sqrt{x})dx.$
3. $\int (2e^x - 5\sin x + 3\sec^2 x)dx.$
4. $\displaystyle\int x^7\,dx$
5. $\displaystyle\int x^{12}\,dx$
6. $\displaystyle\int x^{-3}\,dx$
7. $\displaystyle\int x^{-5}\,dx$
8. $\displaystyle\int x^{1/2}\,dx$
9. $\displaystyle\int x^{3/2}\,dx$
10. $\displaystyle\int x^{-1/2}\,dx$
11. $\displaystyle\int x^{5/3}\,dx$
12. $\displaystyle\int (3x^2 - 4x + 7)\,dx$
13. $\displaystyle\int (5x^4 - 2x^3 + x - 6)\,dx$
14. $\displaystyle\int (2x^5 + 7x^2 - 9)\,dx$
15. $\displaystyle\int (8 - 3x + 4x^3)\,dx$
16. $\displaystyle\int \left(\frac{x^4}{2} - \frac{3x^2}{4} + 5x\right)dx$
17. $\displaystyle\int (6x^7 - 5x^5 + 2x^2 - 1)\,dx$
18. $\displaystyle\int \left(\frac{2}{3}x^6 - \frac{5}{2}x^3 + 4\right)dx$
19. $\displaystyle\int (7x^8 - 3x^4 + x^2 - 8x + 11)\,dx$

Ejercicio 10.2. Radicales y potencias fraccionarias

20. $\displaystyle\int \sqrt{x}\,dx$
21. $\displaystyle\int \sqrt[3]{x^2}\,dx$
22. $\displaystyle\int \frac{1}{\sqrt{x}}\,dx$
23. $\displaystyle\int \frac{1}{\sqrt[3]{x^2}}\,dx$
24. $\displaystyle\int (2\sqrt{x} + 3\sqrt[3]{x})\,dx$
25. $\displaystyle\int \frac{x^2 + \sqrt{x}}{x}\,dx$
26. $\displaystyle\int \left(\sqrt[4]{x^3} + \frac{2}{\sqrt{x}}\right)dx$
27. $\displaystyle\int \frac{x^3 + 2x}{\sqrt{x}}\,dx$

Ejercicio 10.3. Simplificación algebraica

28. $\displaystyle\int \frac{x^3 + 2x^2 - x}{x}\,dx$
29. $\displaystyle\int \frac{3x^4 - 2x^2 + 5x}{x^2}\,dx$

30. $\int \frac{(x+1)^2}{x}\,dx$

31. $\int \frac{x^4-1}{x^2}\,dx$

32. $\int \frac{2x^3+5x^2-3x}{x}\,dx$

33. $\int \frac{x^5+x^3+x}{x^2}\,dx$

34. $\int \frac{(x-2)(x+3)}{x}\,dx$

35. $\int \frac{x^6-2x^3+x}{x^2}\,dx$

36. $\int \frac{x^3+2x^2-x+1}{x^2+1}\,dx$

37. $\int \frac{x^2+3x+5}{x+1}\,dx$

38. $\int \frac{2x^3-x+4}{x^2-1}\,dx$

39. $\int \frac{x^4+1}{x^2+1}\,dx$

40. $\int \frac{3x^3+2x^2+1}{x^2+x}\,dx$

41. $\int \frac{x^3-1}{x-1}\,dx$

42. $\int \frac{x^4-x^2+1}{x^2+1}\,dx$

43. $\int \frac{2x^3+5x^2-3}{x+2}\,dx$

Ejercicio 10.4. Sustitución

44. $\int 2x(x^2+1)^5\,dx$

45. $\int 3x^2(x^3-4)^6\,dx$

46. $\int (2x+1)(x^2+x+3)^4\,dx$

47. $\int 4x^3(x^4+2)^7\,dx$

48. $\int x(3x^2-1)^8\,dx$

49. $\int (6x-5)(3x^2-5x+2)^3\,dx$

50. $\int x^2(1-x^3)^5\,dx$

51. $\int (5x^4+2)(x^5+2x-1)^6\,dx$

52. $\int x(x^2+3)^8 dx.$

53. $\int x^2 e^{x^3} dx.$

54. $\int \frac{5x^4}{x^5+7} dx.$

55. $\int \frac{\cos x}{2+\sin x} dx.$

56. $\int \tan x \sec^2 x dx.$

57. $\int \frac{x}{\sqrt{x^2+4}}\,dx$

58. $\int \frac{2x+1}{\sqrt{x^2+x+5}}\,dx$

59. $\int x^2\sqrt{x^3+1}\,dx$

60. $\int \frac{x^3}{\sqrt{x^4+7}}\,dx$

61. $\int (3x^2-2)\sqrt{x^3-2x+4}\,dx$

62. $\int \frac{4x}{\sqrt[3]{2x^2+1}}\,dx$

63. $\int x^2\sqrt[4]{x^3+2}\,dx$

64. $\int \frac{5x^4}{\sqrt{1+x^5}}\,dx$

Ejercicio 10.5. Por partes

65. $\int xe^{2x}dx.$
66. $\int x\cos 3xdx.$
67. $\int xe^{x}\,dx$
68. $\int x\sin x\,dx$
69. $\int x\cos x\,dx$
70. $\int x\ln x\,dx$
71. $\int x^2e^{x}\,dx$
72. $\int x^2\sin x\,dx$
73. $\int \ln x\,dx$
74. $\int e^{x}\cos x\,dx$
75. $\int x^2\ln xdx.$
76. $\int e^{x}\sin xdx.$
77. $\int \arcsin xdx.$

Ejercicio 10.6. Trigonométricas

78. $\int \sin x\,dx$
79. $\int \cos x\,dx$
80. $\int \sec^2 x\,dx$
81. $\int \csc^2 x\,dx$
82. $\int \sec x\tan x\,dx$
83. $\int \csc x\cot x\,dx$
84. $\int (3\cos x - 2\sin x)\,dx$
85. $\int (4\sec^2 x + 5\csc x\cot x)\,dx$
86. $\int \sin^7 x\cos^2 xdx.$
87. $\int \sin^4 x\cos^2 xdx.$
88. $\int \tan^4 x\sec^4 xdx.$
89. $\int \tan^5 x\sec^3 xdx.$
90. $\int \sin 5x\cos 2xdx.$
91. $\int \sin^2 x\,dx$
92. $\int \cos^2 x\,dx$
93. $\int (1+\cos 2x)\,dx$
94. $\int (1-\cos 2x)\,dx$
95. $\int \sin x\cos x\,dx$
96. $\int \sin(2x)\,dx$
97. $\int \cos(5x)\,dx$
98. $\int (3\sin^2 x + 2\cos^2 x)\,dx$

Ejercicio 10.7. Sustitución trigonométrica

99. $\int \sqrt{16-x^2}dx.$
100. $\int \frac{x^2}{\sqrt{25-x^2}}dx.$
101. $\int \frac{dx}{(x^2+9)^{3/2}}.$
102. $\int \frac{\sqrt{x^2-4}}{x}dx.$
103. $\int \dfrac{dx}{\sqrt{9-x^2}}$

104. $\int \sqrt{16-x^2}\,dx$

105. $\int \dfrac{x^2}{\sqrt{25-x^2}}\,dx$

106. $\int \dfrac{dx}{\sqrt{x^2+9}}$

107. $\int \sqrt{x^2+4}\,dx$

108. $\int \dfrac{x^2}{\sqrt{x^2+16}}\,dx$

109. $\int \dfrac{dx}{\sqrt{x^2-9}}$

110. $\int \dfrac{\sqrt{x^2-25}}{x}\,dx$

Ejercicio 10.8. Fracciones parciales

111. $\int \frac{2x+7}{(x-1)(x+3)}dx.$

112. $\int \frac{x+2}{x(x-1)^2}dx.$

113. $\int \frac{x^2+1}{(x-2)(x^2+4)}dx.$

114. $\int \frac{2x^3+x+1}{(x^2+1)^2}dx.$

115. $\int \frac{x^3+1}{x^2-1}dx$

116. $\int \dfrac{dx}{(x-1)(x+2)}$

117. $\int \dfrac{3x+5}{(x-1)(x+3)}\,dx$

118. $\int \dfrac{dx}{x(x+1)^2}$

119. $\int \dfrac{2x+1}{(x-2)^2(x+1)}\,dx$

120. $\int \dfrac{dx}{(x^2+1)(x-1)}$

121. $\int \dfrac{x^2+1}{x(x-1)(x+2)}\,dx$

122. $\int \dfrac{dx}{(x^2+4)(x^2+1)}$

123. $\int \dfrac{x+1}{(x^2+1)^2}\,dx$

Ejercicio 10.9. Funciones exponenciales

124. $\int e^x\,dx$

125. $\int e^{3x}\,dx$

126. $\int e^{-2x}\,dx$

127. $\int 4e^{5x}\,dx$

128. $\int 2^x\,dx$

129. $\int 5^{3x}\,dx$

130. $\int 3^{-2x}\,dx$

131. $\int (2e^{4x} - 3\,7^x)\,dx$

132. $\int 2^x\,dx.$

133. $\int 3^x\,dx.$

134. $\int 10^x\,dx.$

135. $\int 2^{4x}\,dx.$

136. $\int 3^{2x+1}\,dx.$

137. $\int 5^{3x-2}\,dx.$

138. $\int x\,4^{x^2}\,dx.$

139. $\int x^2\,2^{x^3}\,dx.$

140. $\int (2x+1)\,5^{x^2+x}\,dx.$

141. $\int \cos x\,2^{\sin x}\,dx.$

142. $\int 3^{2x}dx$

143. $\int a^{nx}dx$

144. $\int a^t e^t dt$

145. $\int 5^{x^4+2x}(2x^3+1)dx$

146. $\int x^2 10^{x^3}dx$

147. $\int a^z \ln z(\ln z+1)dz$

148. $\int e^{y^2}e^{3e^y}dy$

149. $\int \frac{4^{\ln(1/x)}}{x}dx$

150. $\int \frac{\log_2 x^2}{x}dx$

151. $\int \frac{(\log_3 x)^2}{x}dx.$

Ejercicio 10.10. Logaritmos

152. $\int \frac{1}{x}\,dx$

153. $\int \frac{5}{x}\,dx$

154. $\int \left(3x^2 + \frac{2}{x}\right) dx$

155. $\int \left(\frac{4}{x} - x^3\right) dx$

156. $\int \frac{7x^2+3}{x}\,dx$

157. $\int \frac{x^3-4}{x}\,dx$

158. $\int \left(\sqrt{x} + \frac{6}{x}\right) dx$

159. $\int \frac{2x^4 - x + 5}{x}\,dx$

160. $\int \frac{1}{3-2x}dx$

161. $\int \frac{x}{2-x^2}dx$

162. $\int \frac{3x^2}{5x^3-1}dx$

163. $\int \frac{2x-1}{x(x-1)}dx$

164. $\int \frac{1}{y \ln y}dy$

165. $\int \frac{\sin 3t}{\cos 3t-1}dt$

166. $\int(\cot 5x + \csc 5x)dx$

167. $\int \frac{\cos 3x+3}{\sin 3x}dx$

168. $\int \frac{2-3\sin 2x}{\cos 2x}dx$

169. $\int(\tan 2x - \sec 2x)dx$

170. $\int \frac{2x^3}{x^2-4}dx$

171. $\int \frac{5-4y^2}{3+2y}dy$

172. $\int \frac{\ln^2(3x)}{x}dx$

173. $\int \frac{2+\ln^2 x}{x(1-\ln x)}dx$

174. $\int \frac{2\ln x+1}{x[(\ln x)^2+\ln x]}dx$

175. $\int \frac{3x^5-2x^3+5x^2-2}{x^3+1}dx$

176. $\int \frac{\tan(\ln x)}{x}dx$

177. $\int \frac{\cot\sqrt{t}}{\sqrt{t}}dt.$

Ejercicio 10.11. $f'(x)/f(x)$

178. $\int \frac{2x}{x^2+3}\,dx$

179. $\int \frac{3x^2}{x^3+5}\,dx$

180. $\int \frac{2x+1}{x^2+x+4}\,dx$

181. $\int \frac{4x^3}{x^4-7}\,dx$

182. $\int \frac{6x-2}{3x^2-2x+9}\,dx$

183. $\int \frac{5x^4+1}{x^5+x+2}\,dx$

184. $\int \frac{\cos x}{2+\sin x}\,dx$

185. $\int \frac{e^x}{1+e^x}\,dx$

Ejercicio 10.12. Sustitución generales

186. $\int e^x(1+e^x)^4\,dx$

187. $\int e^{2x}\sqrt{1+e^{2x}}\,dx$

188. $\int \frac{e^x}{(3+e^x)^2}\,dx$

189. $\int e^{-x}(2+e^{-x})^5\,dx$

190. $\int \frac{3e^{3x}}{1+e^{3x}}\,dx$

191. $\int e^{4x}(1-e^{4x})^3\,dx$

192. $\int \frac{2^x}{1+2^x}\,dx$

193. $\int 5^x(2+5^x)^6\,dx$

194. $\int \sin x\cos^4 x\,dx$

195. $\int \cos x\sin^5 x\,dx$

196. $\int \sec^2 x(1+\tan x)^3\,dx$

197. $\int \csc^2 x(2-\cot x)^4\,dx$

198. $\int \sec x\tan x(3+\sec x)^5\,dx$

199. $\int \csc x\cot x(1+\csc x)^2\,dx$

200. $\int \frac{\cos x}{1+\sin x}\,dx$

201. $\int \frac{\sec^2 x}{2+\tan x}\,dx$

Ejercicio 10.13. Trigonométricas inversas

202. $\int \frac{1}{1+x^2}\,dx$

203. $\int \frac{1}{\sqrt{1-x^2}}\,dx$

204. $\int \frac{1}{4+x^2}\,dx$

205. $\int \frac{1}{\sqrt{9-x^2}}\,dx$

206. $\int \frac{1}{1+4x^2}\,dx$

207. $\int \frac{1}{\sqrt{16-9x^2}}\,dx$

208. $\int \frac{3}{1+9x^2}\,dx$

209. $\int \frac{2}{\sqrt{25-4x^2}}\,dx$

210. $\int \frac{dx}{\sqrt{1-x^2}}$

211. $\int \frac{dx}{\sqrt{4-x^2}}$

212. $\int \frac{dx}{\sqrt{25-4x^2}}$

213. $\int \frac{3\,dx}{\sqrt{16-9x^2}}$

214. $\int \frac{dx}{1+x^2}$

215. $\int \frac{dx}{9+x^2}$

216. $\int \frac{dx}{4+9x^2}$

217. $\int \frac{5\,dx}{25+4x^2}$

Ejercicio 10.14. Potencias de trigonométricas

218. $\int \sin^3 x\,dx$

219. $\int \sin^5 x\,dx$

220. $\int \sin^7 x\,dx$

221. $\int \sin^2 x\,dx$

222. $\int \sin^4 x\,dx$

223. $\int \sin^6 x\,dx$

224. $\int \sin^3(2x)\,dx$

225. $\int \sin^4(3x)\,dx$

226. $\int \cos^3 x\,dx$

227. $\int \cos^5 x\,dx$

228. $\int \cos^7 x\,dx$

229. $\int \cos^2 x\,dx$

230. $\int \cos^4 x\,dx$

231. $\int \cos^6 x\,dx$

232. $\int \cos^3(2x)\,dx$

233. $\int \cos^4(3x)\,dx$

234. $\int \sin(3x)\cos(5x)\,dx$

235. $\int \cos(2x)\cos(7x)\,dx$

236. $\int \sin(4x)\sin(2x)\,dx$

237. $\int \sin x\cos(3x)\,dx$

238. $\int \cos(5x)\cos x\,dx$

239. $\int \sin(6x)\sin(3x)\,dx$

240. $\int \sin(2x)\cos(4x)\,dx$

241. $\int \cos(3x)\sin(7x)\,dx$

242. $\int \sin^3 x\cos^2 x\,dx$

243. $\int \sin^2 x\cos^3 x\,dx$

244. $\int \sin^5 x\cos^4 x\,dx$

245. $\int \sin^4 x\cos^5 x\,dx$

246. $\int \sin^2 x\cos^2 x\,dx$

247. $\int \sin^4 x\cos^2 x\,dx$

248. $\int \sin^2 x\cos^4 x\,dx$

249. $\int \sin^4 x\cos^4 x\,dx$

250. $\int \tan^3 x\,dx$

251. $\int \sec^3 x\,dx$

252. $\int \tan^2 x\sec^2 x\,dx$

253. $\int \tan^4 x\sec^2 x\,dx$

254. $\int \sec^4 x\tan x\,dx$

255. $\int \sec^2 x\tan^3 x\,dx$

256. $\int \tan^4 x\,dx$

257. $\int \sec^5 x\,dx$

258. $\int \cot^3 x\, dx$

259. $\int \csc^3 x\, dx$

260. $\int \cot^2 x \csc^2 x\, dx$

261. $\int \cot^4 x \csc^2 x\, dx$

262. $\int \csc^4 x \cot x\, dx$

263. $\int \csc^2 x \cot^3 x\, dx$

264. $\int \cot^4 x\, dx$

265. $\int \csc^5 x\, dx$

Ejercicio 10.15. Identificar el método

266. $\int (4x^3 - 2x + 7)\, dx$

267. $\int \dfrac{x^2 + 3x}{x}\, dx$

268. $\int \dfrac{x}{x^2 + 9}\, dx$

269. $\int x^2(x^3 + 4)^7\, dx$

270. $\int \dfrac{x}{\sqrt{x^2 + 1}}\, dx$

271. $\int e^{2x}(1 + e^{2x})^3\, dx$

272. $\int \cos x(2 + \sin x)^5\, dx$

273. $\int \dfrac{\sec^2 x}{1 + \tan x}\, dx$

274. $\int \cos^2 x\, dx$

275. $\int \dfrac{1}{9 + x^2}\, dx$

276. $\int \dfrac{1}{\sqrt{25 - x^2}}\, dx$

277. $\int \left(x^{3/2} + x^{-1/2}\right) dx$

278. $\int 7^{2x}\, dx$

279. $\int \dfrac{4x^3}{2 + x^4}\, dx$

280. $\int \sin x \cos^6 x\, dx$

281. $\int x \arctan x\, dx$

282. $\int x^2 \ln x\, dx$

283. $\int xe^{x^2} \cos(e^{x^2})\, dx$

284. $\int \dfrac{x^3}{\sqrt{9 - x^2}}\, dx$

285. $\int \dfrac{x^2}{x^2 + 4}\, dx$

286. $\int e^x \sin x\, dx$

287. $\int \dfrac{x^2 + 1}{x^3 + x}\, dx$

288. $\int \sin^3 x \cos^4 x\, dx$

289. $\int \dfrac{e^x}{4 + e^x}\, dx$

290. $\int \dfrac{x^5 + 2x^3 - x}{x^2}\, dx$

291. $\int \dfrac{x^2}{\sqrt{1 + x^3}}\, dx$

292. $\displaystyle\int \frac{2x-3}{x^2-3x+10}\,dx$

293. $\displaystyle\int e^{-3x}(1+e^{-3x})^7\,dx$

294. $\displaystyle\int \frac{\cos(2x)}{3+\sin(2x)}\,dx$

295. $\displaystyle\int \frac{1}{16+25x^2}\,dx$

296. $\displaystyle\int \frac{1}{\sqrt{36-4x^2}}\,dx$

297. $\displaystyle\int \left(2\sin^2 x+3\cos^2 x\right)\,dx$

298. $\int 3x^4dx$

299. $\int 2x^7dx$

300. $\int x^{-3}dx$

301. $\int 3t^{-5}dt$

302. $\int 5u^{3/2}du$

303. $\int 10\sqrt[3]{x^2}dx$

304. $\int \frac{2}{\sqrt[3]{x}}dx$

305. $\int \frac{3}{\sqrt{y}}dy$

306. $\int 6t^2\sqrt[3]{t}\,dt$

307. $\int(3u^5-2u^3)du$

308. $\int y^3(2y^2-3)dy$

309. $\int x^4(5-x^2)dx$

310. $\int(8x^4+4x^3-6x^2-4x+5)dx$

311. $\int(2+3x^2-8x^3)dx$

312. $\int \sqrt{x}(x+1)dx$

313. $\int(\sqrt{x}-1/\sqrt{x})dx$

314. $\int(2/x^3+3/x^2+5)dx$

315. $\int(3-1/x^4+1/x^2)dx$

316. $\int \frac{x^2+4x-4}{\sqrt{x}}dx$

317. $\int \frac{y^4+2y^2-1}{\sqrt{y}}dy$

318. $\int(\sqrt[3]{x}+1/\sqrt[3]{x})dx$

319. $\int \frac{27t^3-1}{\sqrt[3]{t}}dt$

320. $\int(3\sin t-2\cos t)dt$

321. $\int(5\cos x-4\sin x)dx$

322. $\int \frac{\sin x}{\cos^2 x}dx$

323. $\int \frac{\cos x}{\sin^2 x}dx$

324. $\int(4\csc x\cot x+2\sec^2 x)dx$

325. $\int(3\csc^2 t-5\sec t\tan t)dt.$

326. $\int \sqrt{1-4y}\,dy$

327. $\int \sqrt[3]{3x-4}\,dx$

328. $\int x\sqrt[3]{x^2-9}\,dx$

329. $\int x(2x^2+1)^6dx$

330. $\int x^2(x^3-1)^{10}dx$

331. $\int 3x\sqrt{4-x^2}dx$

332. $\int \frac{y^3}{(1-2y^4)^5}dy$

333. $\int \frac{s}{\sqrt{3s^2+1}}ds$

334. $\int(x^2-4x+4)^{4/3}dx$

335. $\int x^4\sqrt{3x^5-5}dx$

336. $\int x\sqrt{x+2}dx$

337. $\int \frac{t}{\sqrt{t+3}}dt$

338. $\int \frac{2r}{(1-r)^7}dr$

339. $\int x^3(2-x^2)^{12}dx$

340. $\int x^2\sqrt{3-2x}dx$

341. $\int(r^3+3)^{1/4}r^5dr$

342. $\int \cos 4\theta\,d\theta$

343. $\int \sin(x/3)dx$

344. $\int 6x^2 \sin(x^3)dx$

345. $\int \frac{1}{2}t\cos(4t^2)dt$

346. $\int \sec^2(5x)dx$

347. $\int \csc^2(2\theta)d\theta$

348. $\int y\csc(3y^2)\cot(3y^2)dy$

349. $\int r^2\sec^2(r^3)dr$

350. $\int \cos x(2+\sin x)^4 dx$

351. $\int \frac{4\sin x}{(1+\cos x)^2}dx$

352. $\int \sqrt{1+\frac{1}{3x}}\frac{1}{x^2}dx$

353. $\int \sqrt{\frac{1}{t}-1}\frac{1}{t^2}dt$

354. $\int 2\sin x\sqrt[3]{1+\cos x}dx$

355. $\int \sin 2x\sqrt{2-\cos 2x}dx$

356. $\int \cos^2 t\sin t\,dt$

357. $\int \sin^3\theta\cos\theta\,d\theta$

358. $\int(\tan 2x+\cot 2x)^2dx$

359. $\int \frac{\sec^2(3\sqrt{t})}{\sqrt{t}}dt$

360. $\int \frac{x^2+2x}{\sqrt{x^3+3x^2+1}}dx$

361. $\int x(x^2+1)\sqrt{4-2x^2-x^4}dx$

362. $\int \frac{y+3}{(3-y)^{2/3}}dy$

363. $\int \sqrt{3+s}(s+1)^2 ds$

364. $\int \frac{(r^{1/3}+2)^4}{\sqrt[3]{r^2}}dr$

365. $\int (t+1/t)^{3/2}\frac{t^2-1}{t^2}dt$

366. $\int \frac{x^3}{(x^2+4)^{3/2}}dx$

367. $\int \frac{x^3}{\sqrt{1-2x^2}}dx$

368. $\int \sin x\sin(\cos x)dx$

369. $\int \sec x\tan x\cos(\sec x)dx.$

370. $\int \frac{dx}{\sqrt{36-x^2}}.$

371. $\int \frac{dx}{16+x^2}.$

372. $\int \frac{dx}{\sqrt{9-x^2}}.$

373. $\int \frac{dx}{25+x^2}.$

374. $\int \frac{dx}{\sqrt{25-4x^2}}.$

375. $\int \frac{dx}{9+16x^2}.$

376. $\int \frac{5\,dx}{\sqrt{49-25x^2}}.$

377. $\int \frac{6\,dx}{4+9x^2}.$

378. $\int \frac{2x\,dx}{1+x^4}.$

379. $\int \frac{3x^2\,dx}{\sqrt{1-x^6}}.$

380. $\int \frac{e^x\,dx}{1+e^{2x}}.$

381. $\int \frac{\cos x\,dx}{\sqrt{1-\sin^2 x}},$

382. $\int \frac{dx}{x^2+4x+8}.$

383. $\int \frac{dx}{x^2-6x+18}.$

384. $\int \frac{dx}{2x^2+8x+10}.$

385. $\int \frac{dx}{4x^2-8x+8}.$

386. $\int \frac{dx}{\sqrt{1-4x^2}}$

387. $\int \frac{dx}{x^2+25}$

388. $\int \frac{dx}{9x^2+16}$

389. $\int \frac{dt}{\sqrt{1-16t^2}}$

390. $\int \frac{dx}{4x\sqrt{x^2-16}}$

391. $\int \frac{x}{x^4+16}dx$

392. $\int \frac{r}{\sqrt{16-9r^4}}dr$

393. $\int \frac{du}{u\sqrt{16u^2-9}}$

394. $\int \frac{e^x}{7+e^{2x}}dx$

395. $\int \frac{\sin x}{\sqrt{2-\cos^2 x}}dx$

396. $\int \frac{dx}{(1+x)\sqrt{x}}$

397. $\int \frac{ds}{\sqrt{2s-s^2}}$

398. $\int \frac{dx}{x^2-x+2}$

399. $\int \frac{dx}{\sqrt{3x-x^2-2}}$

400. $\int \frac{dx}{\sqrt{15+2x-x^2}}$

401. $\int \frac{2dt}{(t-3)\sqrt{t^2-6t+5}}$.

402. $\displaystyle\int \frac{x^2}{\sqrt{25-x^2}}\,dx.$

403. $\displaystyle\int \frac{dx}{(x^2+9)^{3/2}}.$

404. $\displaystyle\int \frac{\sqrt{x^2-16}}{x}\,dx.$

405. $\displaystyle\int \frac{dx}{\sqrt{36-x^2}}.$

406. $\displaystyle\int \sqrt{x^2+4}\,dx.$

407. $\displaystyle\int \frac{x^2}{\sqrt{x^2-9}}\,dx.$

408. $\int(3x^5-4x^2+1)dx.$

409. $\int x(x^2+5)^6dx.$

410. $\int \frac{x}{x^2+4}dx.$

411. $\int xe^x dx.$

412. $\int x^2\cos x dx.$

413. $\int \sin^3 x\cos^4 x dx.$

414. $\int \sin^2 x\cos^2 x dx.$

415. $\int \sqrt{25-x^2}dx.$

416. $\int \frac{2x+3}{(x-1)(x+2)}dx.$

417. $\int \frac{x^2+3x+4}{x+1}dx.$

418. $\int \ln x dx.$

419. $\int xe^{x^2}dx.$

420. $\int \frac{3x^2}{x^3+1}dx.$

421. $\int x^2e^x dx.$

422. $\int \frac{dx}{\sqrt{9-x^2}}.$

423. $\int \frac{x^4}{x^2+1}dx.$

424. $\int \cos 3x\cos 7x dx.$

425. $\int \frac{x}{(x^2+1)^2}dx.$

426. $\int \sec^3 x dx.$

427. $\int e^x\cos x dx.$

Formulario de integración

C constante; $a > 0$, $a \neq 1$; $k \neq 0$ cuando aparece en denominadores

Algebraicas y logarítmicas

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$
$$\int \frac{dx}{x} = \ln|x| + C$$
$$\int \frac{u'}{u}\,dx = \ln|u| + C$$
$$\int \ln x\,dx = x\ln x - x + C$$
$$\int (\alpha f + \beta g)\,dx = \alpha \int f\,dx + \beta \int g\,dx$$

Exponenciales

$$a^x = e^{x\ln a}$$
$$\int e^{kx}\,dx = \frac{e^{kx}}{k} + C$$
$$\int a^{kx}\,dx = \frac{a^{kx}}{k\ln a} + C$$
$$\int e^u u'\,dx = e^u + C$$
$$\int a^u u'\,dx = \frac{a^u}{\ln a} + C$$

Trigonométricas

$$\int \sin x\,dx = -\cos x + C$$
$$\int \cos x\,dx = \sin x + C$$
$$\int \sec^2 x\,dx = \tan x + C$$
$$\int \csc^2 x\,dx = -\cot x + C$$
$$\int \sec x\tan x\,dx = \sec x + C$$
$$\int \csc x\cot x\,dx = -\csc x + C$$
$$\int \tan x\,dx = -\ln|\cos x| + C$$
$$\int \cot x\,dx = \ln|\sin x| + C$$
$$\int \sec x\,dx = \ln|\sec x + \tan x| + C$$
$$\int \csc x\,dx = \ln|\csc x - \cot x| + C$$

Identidades trigonométricas

$$\sin^2 x + \cos^2 x = 1$$
$$1 + \tan^2 x = \sec^2 x$$
$$1 + \cot^2 x = \csc^2 x$$
$$\sin 2x = 2\sin x\cos x$$
$$\cos 2x = \cos^2 x - \sin^2 x$$
$$\sin^2 x = \frac{1-\cos 2x}{2}$$
$$\cos^2 x = \frac{1+\cos 2x}{2}$$

Logaritmos

$$\log_a(MN) = \log_a M + \log_a N$$
$$\log_a \frac{M}{N} = \log_a M - \log_a N$$
$$\log_a(M^r) = r\log_a M$$
$$\log_a x = \frac{\ln x}{\ln a}$$

Inversas trigonométricas

$$\frac{d}{dx}\arcsin u = \frac{u'}{\sqrt{1-u^2}}$$
$$\frac{d}{dx}\arccos u = -\frac{u'}{\sqrt{1-u^2}}$$
$$\frac{d}{dx}\arctan u = \frac{u'}{1+u^2}$$
$$\frac{d}{dx}\operatorname{arccot} u = -\frac{u'}{1+u^2}$$
$$\frac{d}{dx}\operatorname{arcsec} u = \frac{u'}{|u|\sqrt{u^2-1}}$$
$$\frac{d}{dx}\operatorname{arccsc} u = -\frac{u'}{|u|\sqrt{u^2-1}}$$

Integrales con inversas

$$\int \frac{du}{\sqrt{1-u^2}} = \arcsin u + C$$
$$\int \frac{du}{1+u^2} = \arctan u + C$$
$$\int \frac{du}{|u|\sqrt{u^2-1}} = \operatorname{arcsec}|u| + C$$
$$\int \frac{du}{\sqrt{a^2-u^2}} = \arcsin\frac{u}{a} + C$$
$$\int \frac{du}{a^2+u^2} = \frac{1}{a}\arctan\frac{u}{a} + C$$
$$\int \frac{du}{|u|\sqrt{u^2-a^2}} = \frac{1}{a}\operatorname{arcsec}\frac{|u|}{a} + C$$

Métodos de integración

$$u = g(x), \qquad du = g'(x)\,dx$$
$$\int f(g(x))g'(x)\,dx = \int f(u)\,du$$
$$\int u\,dv = uv - \int v\,du$$

Sustituciones trigonométricas

$$\sqrt{a^2-x^2}: \quad x = a\sin\theta$$
$$dx = a\cos\theta\,d\theta$$
$$\sqrt{a^2+x^2}: \quad x = a\tan\theta$$
$$dx = a\sec^2\theta\,d\theta$$
$$\sqrt{x^2-a^2}: \quad x = a\sec\theta$$
$$dx = a\sec\theta\tan\theta\,d\theta$$